

Quantitative Aptitude: Number System

Complete Study Guide, Solved Examples & 25 Practice Questions

1. Theory & Classification of Numbers

The Number System is a fundamental branch of mathematics that deals with different types of numbers, their properties, and mathematical operations. It is one of the most vital topics for competitive examinations like SSC, Banking, Railways, and CAT.

Classification of Numbers:

- **Natural Numbers (N):** Counting numbers starting from 1. $N = \{1, 2, 3, 4, \dots\}$
- **Whole Numbers (W):** Natural numbers including zero (0). $W = \{0, 1, 2, 3, \dots\}$
- **Integers (Z):** All positive numbers, negative numbers, and zero. $Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- **Even Numbers:** Numbers completely divisible by 2 (e.g., 2, 4, 6, 8, $\dots$).
- **Odd Numbers:** Numbers not divisible by 2 (e.g., 1, 3, 5, 7, $\dots$).
- **Prime Numbers:** Numbers having only two factors: 1 and itself (e.g., 2, 3, 5, 7, 11, 13, 17, 19). 2 is the smallest and only even prime number.
- **Composite Numbers:** Numbers having more than two factors (e.g., 4, 6, 8, 9, 10).
- **Rational Numbers:** Numbers that can be expressed in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$.
- **Irrational Numbers:** Numbers that cannot be expressed in $\frac{p}{q}$ form (e.g., $\sqrt{2}$, π , e).

Important Formulas:

- Sum of first n natural numbers = $\frac{n(n+1)}{2}$
- Sum of first n even numbers = $n(n+1)$
- Sum of first n odd numbers = n^2
- Sum of squares of first n natural numbers = $\frac{n(n+1)(2n+1)}{6}$
- Sum of cubes of first n natural numbers = $\left[\frac{n(n+1)}{2} \right]^2$

2. Solved Examples (10 Problems)

Example 1: Find the sum of the first 20 natural numbers.

Solution: Formula = $\frac{n(n+1)}{2}$, Here $n = 20$

Sum = $\frac{20 \times (20+1)}{2} = \frac{20 \times 21}{2} = 10 \times 21 = 210$.

Example 2: Find the sum of the first 15 even numbers.

Solution: Formula = $n(n+1)$, Here $n = 15$

Sum = $15 \times (15+1) = 15 \times 16 = 240$.

Example 3: What is the sum of all odd numbers from 1 to 25?

Solution: Total odd numbers from 1 to 25 is $n = \frac{25+1}{2} = 13$.

Sum = $n^2 = (13)^2 = 169$.

Example 4: What digit should replace the star (*) in the number 47392^* so that it is completely divisible by 3?

Solution: Divisibility rule for 3: The sum of the digits must be divisible by 3.

Sum of digits = $4 + 7 + 3 + 9 + 2 = 25$.

The next multiple of 3 after 25 is 27. Thus, $27 - 25 = 2$. The missing digit is $^{**}2^{**}$.

Example 5: If the number $978x5$ is divisible by 11, find the value of x .

Solution: Divisibility rule for 11: The difference between the sum of digits at odd places and even places must be 0 or a multiple of 11.

Sum of digits at odd places (from right): $5 + 8 + 9 = 22$

Sum of digits at even places: $x + 7$

Difference = $22 - (7 + x) = 15 - x$. For this to be equal to 11, $15 - x = 11 \implies x = 4$.

Example 6: Find the unit digit of the expression $(7^{95} - 3^{58})$.

Solution:

1) 7^{95} : Cyclicity of 7 is 4. $95 \div 4$ leaves a remainder of 3. Unit digit = $7^3 = 343$ (digit is 3).

2) 3^{58} : Cyclicity of 3 is 4. $58 \div 4$ leaves a remainder of 2. Unit digit = $3^2 = 9$.

Unit digit = $3 - 9$. Borrowing 1 from the tens place: $13 - 9 = 4$.

Example 7: The product of two numbers is 120 and the sum of their squares is 289. Find the sum of the two numbers.

Solution: Let the numbers be a and b . $ab = 120$ and $a^2 + b^2 = 289$.

We know that $(a+b)^2 = a^2 + b^2 + 2ab$

$$(a+b)^2 = 289 + 2(120) = 289 + 240 = 529$$

$$a + b = \sqrt{529} = 23.$$

Example 8: A number when divided by 56 leaves a remainder of 29. If the same number is divided by 8, what will be the remainder?

Solution: Divide the previous remainder directly by the new divisor.

New remainder = $29 \div 8$ remainder = 5 ($8 \times 3 = 24$, remainder $29 - 24 = 5$).

Example 9: How many prime numbers are there between 1 and 100?

Solution: There are exactly 25 prime numbers between 1 and 100.

Example 10: Find the value of $1^2 + 2^2 + 3^2 + \dots + 10^2$.

Solution: Formula = $\frac{n(n+1)(2n+1)}{6}$, Here $n = 10$

$$\text{Value} = \frac{10 \times (10+1) \times (2 \times 10+1)}{6} = \frac{10 \times 11 \times 21}{6} = \frac{2310}{6} = 385.$$

3. Practice Questions (25 Questions)

Q1. What is the sum of all natural numbers from 1 to 50?

- (A) 1275 (B) 1250
(C) 1320 (D) 1225
-

Q2. Find the sum of the first 20 even numbers.

- (A) 400 (B) 420
(C) 380 (D) 440
-

Q3. What is the sum of all odd numbers from 1 to 40?

- (A) 1600 (B) 800
(C) 840 (D) 780
-

Q4. Which least digit should replace (*) in $5824*$ so that the number is completely divisible by 4?

- (A) 0 (B) 2
(C) 4 (D) 6
-

Q5. If the number $725*3$ is divisible by 9, find the value of (*).

- (A) 1 (B) 3
(C) 5 (D) 7
-

Q6. What is the unit digit in the product $(2467)^{153} \times (341)^{72}$?

- (A) 7 (B) 3
(C) 1 (D) 9
-

Q7. A number when divided by 119 leaves a remainder of 19. What will be the remainder if it is divided by 17?

- (A) 0 (B) 2
(C) 3 (D) 5
-

Q8. The sum of three consecutive natural numbers is 87. Find the middle number.

- (A) 27 (B) 28
(C) 29 (D) 30
-

Q9. Find the value of $11^2 + 12^2 + 13^2 + \dots + 20^2$.

- (A) 2485 (B) 2870
(C) 385 (D) 2255
-

Q10. Which is the smallest prime number?

- (A) 0 (B) 1
(C) 2 (D) 3
-

Q11. How many prime numbers are there less than 50?

- (A) 15 (B) 16
(C) 17 (D) 18
-

Q12. The sum of two numbers is 40 and their difference is 6. Find the product of the two numbers.

- (A) 391 (B) 385
(C) 371 (D) 400
-

Q13. What is the unit digit of 7^{105} ?

- (A) 1 (B) 3
(C) 7 (D) 9
-

Q14. If $\frac{3}{5}$ of a number is 10 more than $\frac{1}{2}$ of the same number, what is the number?

- (A) 50 (B) 60
(C) 80 (D) 100
-

Q15. How many numbers between 100 and 200 are completely divisible by 3?

- (A) 32 (B) 33
(C) 34 (D) 35
-

Q16. A number when divided by 361 leaves a remainder of 47. What will be the remainder if the same number is divided by 19?

(A) 8

(B) 9

(C) 10

(D) 11

Q17. The product of two numbers is 336. Their sum exceeds their difference by 28. What are the numbers?

(A) 24, 14

(B) 21, 16

(C) 28, 12

(D) 24, 12

Q18. Find the value of $1^3 + 2^3 + 3^3 + \dots + 9^3$.

(A) 2025

(B) 2125

(C) 1980

(D) 2250

Q19. Which of the following numbers is a prime number?

(A) 91

(B) 117

(C) 101

(D) 143

Q20. If n is an even integer, then $(n^3 - n)$ is always divisible by:

(A) 3

(B) 6

(C) 12

(D) None of these

Q21. What is the sum of all numbers from 1 to 60 that are divisible by 3?

(A) 630

(B) 590

(C) 610

(D) 650

Q22. Subtracting 10 from a positive number yields 75 times its reciprocal. Find the number.

(A) 12

(B) 15

(C) 20

(D) 25

Q23. What will be the remainder when $(8^{19} + 1)$ is divided by 9?

(A) 0

(B) 1

(C) 2

(D) 8

Q24. What is the sum of prime numbers between 50 and 80?

- (A) 321 (B) 351
(C) 311 (D) 341

Q25. The sum of a two-digit number and the number formed by reversing its digits is 132. If the difference between the digits is 4, find the number.

- (A) 84 (B) 73
(C) 95 (D) 62

4. Answer Key

Correct answers for all 25 practice questions:

Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer
Q1	(D) 1225	Q8	(C) 29	Q15	(B) 33	Q22	(B) 15
Q2	(B) 420	Q9	(A) 2485	Q16	(A) 8	Q23	(A) 0
Q3	(B) 800	Q10	(C) 2	Q17	(A) 24, 14	Q24	(B) 351
Q4	(A) 0	Q11	(A) 15	Q18	(A) 2025	Q25	(A) 84
Q5	(A) 1	Q12	(C) 371	Q19	(C) 101		
Q6	(B) 3	Q13	(B) 3	Q20	(B) 6		
Q7	(C) 3	Q14	(D) 100	Q21	(A) 630		